Integration by Parts

ClipSAT Printable Practice Suite (clipsat.org) • Topic 7.02

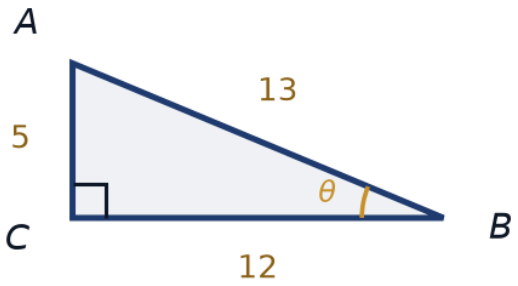
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: The integration by parts formula

Evaluate $\int xe^x\,dx$ using integration by parts.

2 [Foundational] • Target: The integration by parts formula

Evaluate $\int x\cos x\,dx$ using integration by parts.



3 [Foundational] • Target: The integration by parts formula

Evaluate $\int \ln x\,dx$ using integration by parts (with $dv = dx$).

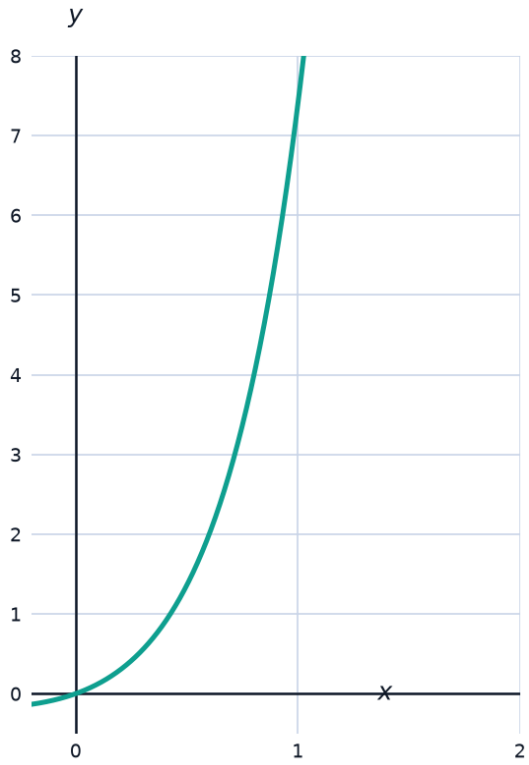
4 [Foundational] • Target: Repeated and definite applications

Evaluate $\int x^2e^x\,dx$, applying integration by parts twice.

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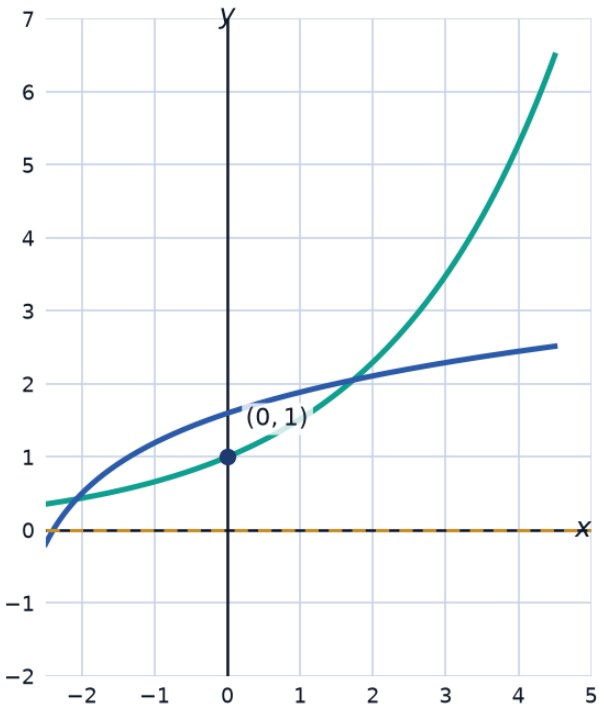
5 [Foundational] • Target: Repeated and definite applications

Evaluate $\int_0^1 xe^{2x}\,dx$ exactly.



6 [Foundational] • Target: Choosing u and dv wisely (LIATE)

Evaluate $\int x\ln x\,dx$, choosing $u = \ln x$ (logarithmic functions are prioritized as u under the LIATE rule).



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7 [Foundational] • Target: Choosing u and dv wisely (LIATE)

Evaluate $\int x^2 \cos x \, dx$, applying integration by parts twice.

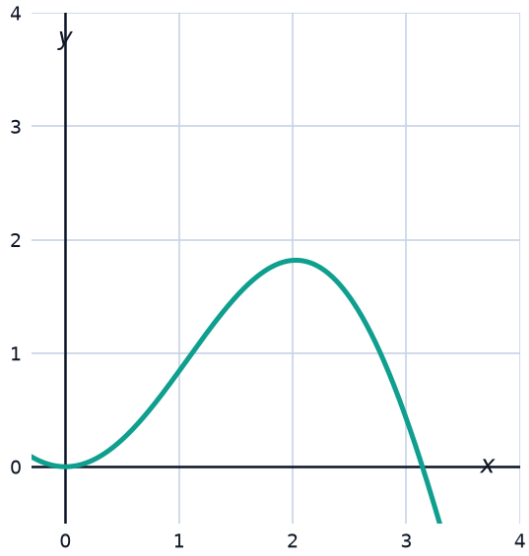
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: The recurring-integral trick

Evaluate $\int e^x \sin x \, dx$ using the technique of applying integration by parts twice and solving algebraically for the original integral.

9 [Intermediate] • Target: Definite integrals by parts

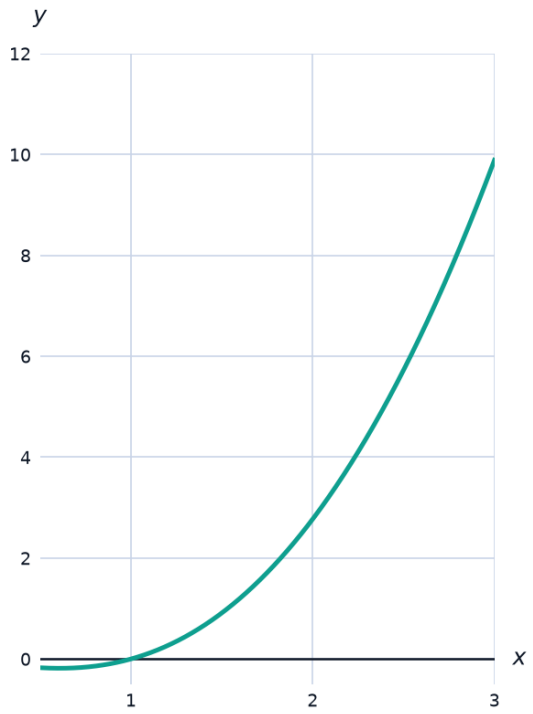
Evaluate $\int_0^{\pi} x \sin x \, dx$ exactly.



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10 [Intermediate] • Target: Definite integrals by parts

Evaluate $\int_1^e x^2 \ln x \, dx$ exactly.



11 [Intermediate] • Target: Reduction formulas

Use integration by parts with $u = x^n$, $dv = e^x dx$ to derive the reduction formula $\int x^n e^x \, dx = x^n e^x - n\int x^{n-1} e^x \, dx$.

12 [Intermediate] • Target: Reduction formulas

Use the reduction formula above (applied twice) to evaluate $\int x^2 e^x \, dx$, confirming it matches an earlier result.

13 [Intermediate] • Target: Integration by parts with inverse trig functions

Evaluate $\int \tan^{-1} x \, dx$ using integration by parts with $u = \tan^{-1} x$, $dv = dx$.

14 [Intermediate] • Target: Integration by parts with inverse trig functions

Evaluate $\int x \sin^{-1} x \, dx$ using $u = \sin^{-1} x$, $dv = x \, dx$ (the resulting integral can be left in terms of a known antiderivative form).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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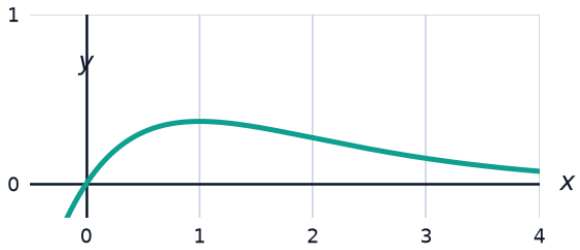
15 **[Challenging]** • **Target: A capstone mixed-technique problem** • **Cross-Lesson: 2.02 Continuity**

Evaluate $\int_0^{\pi/2} x\cos(2x)\,dx$ exactly, using integration by parts.



16 **[Challenging]** • **Target: Visualizing an area found by integration by parts** • **Cross-Lesson: 3.01 The Derivative from First Principles**

The graph shows $f(x) = xe^{-x}$ on $[0, 3]$. Find the exact area under the curve on this interval using integration by parts.



17 **[Challenging]** • **Target: Choosing between substitution and parts** • **Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain**

For each integral, state whether substitution or integration by parts is the appropriate technique, and identify the key choice (u-substitution variable, or u/dv for parts) without fully evaluating: (a) $\int xe^{x^2}\,dx$; (b) $\int xe^{2x}\,dx$.

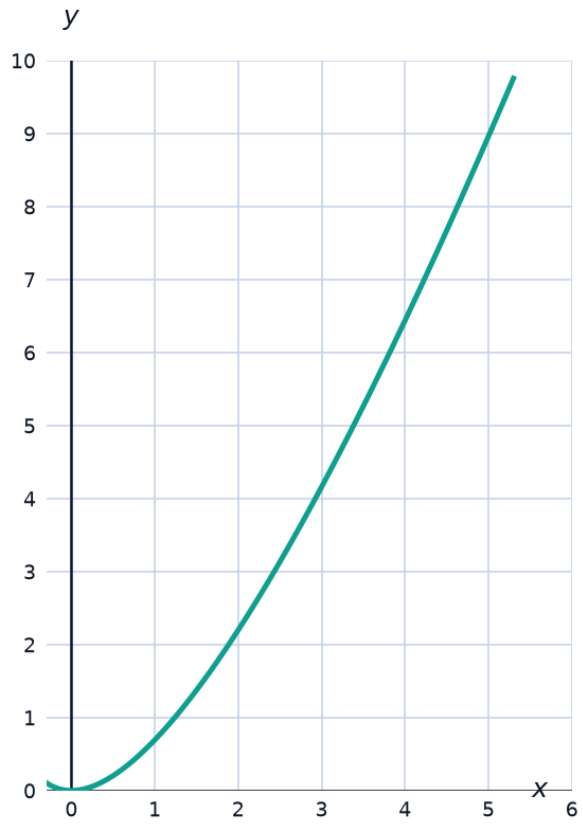
18 **[Challenging]** • **Target: Choosing between substitution and parts** • **Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals**

Evaluate $\int \sec^3 x\,dx$ using integration by parts with $u = \sec x$, $dv = \sec^2 x\,dx$, and the identity $\tan^2 x = \sec^2 x - 1$ to solve the resulting equation for the original integral.

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19 [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

A city's population growth rate is modeled by $r(t) = t\ln(t+1)$ thousand people per year, for $0 \le t \le 5$. Set up and evaluate the definite integral for the total population growth over these 5 years using integration by parts, giving your answer to two decimal places.



20 [Challenging] • Target: Reduction formulas • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Area between $y = x$ and $y = x^2$: intersect at $x =$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Standard Products (Polynomial × Exponential / Trig)

State the formula for integration by parts.

a. $\int u\,dv = uv - \int v\,du$

b. $\int u\,dv = uv + \int v\,du$

c. $\int u\,dv = u'v - uv'$

d. $\int u\,dv = \frac{u}{v} - \int u'v$

22 [Foundational] • Target: Standard Products (Polynomial × Exponential / Trig)

Evaluate $\int xe^{2x}\,dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$

b. $\frac{1}{2}xe^{2x} - \frac{1}{2}e^{2x} + C$

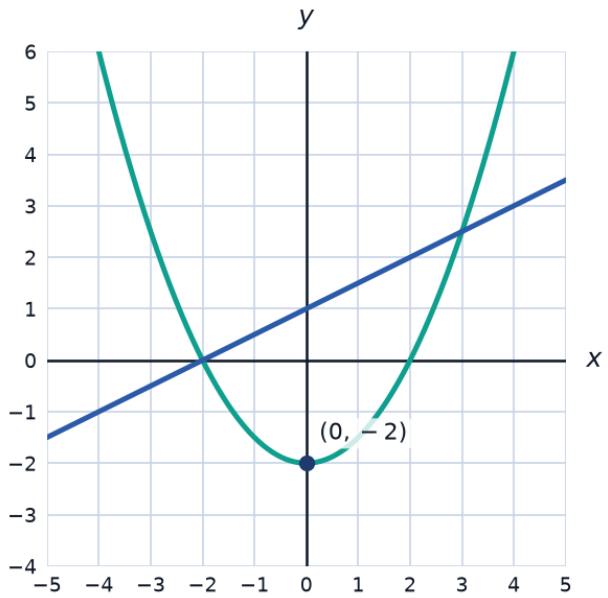
c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{4}xe^{2x} - \frac{1}{4}e^{2x} + C$

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23 [Foundational] • Target: Standard Products (Polynomial × Exponential / Trig)

Evaluate $\int x\sin(x)\,dx$.



a. $-x\cos x + \sin x + C$

b. $x\cos x - \sin x + C$

c. $-x\cos x - \sin x + C$

d. $x\sin x + \cos x + C$

24 [Foundational] • Target: Logarithmic and Inverse Trig Integrands

Evaluate $\int \ln(x)\,dx$.

a. $x\ln x - x + C$

b. $\frac{1}{x} + C$

c. $x\ln x + x + C$

d. $\frac{(\ln x)^2}{2} + C$

25 [Foundational] • Target: Logarithmic and Inverse Trig Integrands

Evaluate $\int \arctan(x)\,dx$.

a. $x\arctan x - \frac{1}{2}\ln(1 + x^2) + C$

b. $x\arctan x - \ln(1 + x^2) + C$

c. $\frac{1}{1 + x^2} + C$

d. $x\arctan x + \frac{1}{2}\ln(1 + x^2) + C$

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26 [Foundational] • Target: Logarithmic and Inverse Trig Integrands

Evaluate $\int x\ln(x)\,dx$.

a. $\frac{x^2}{2}\ln x - \frac{x^2}{4} + C$

b. $\frac{x^2}{2}\ln x - \frac{x^2}{2} + C$

c. $x^2\ln x - x^2 + C$

d. $\frac{x^2}{2}\ln x + \frac{x^2}{4} + C$

27 [Foundational] • Target: Repeated Integration and Tabular Method

Evaluate $\int x^2 e^x\,dx$.

a. $x^2e^x - 2xe^x + 2e^x + C$

b. $x^2e^x - 2xe^x + C$

c. $x^2e^x + 2xe^x + 2e^x + C$

d. $\frac{x^3}{3}e^x + C$

28 [Foundational] • Target: Repeated Integration and Tabular Method

Evaluate $\int x^2\cos(x)\,dx$.

a. $x^2\sin x + 2x\cos x - 2\sin x + C$

b. $x^2\sin x - 2x\cos x + 2\sin x + C$

c. $-x^2\cos x + 2x\sin x + C$

d. $x^2\sin x + 2x\cos x + C$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Repeated Integration and Tabular Method

Evaluate $\int_0^1 x^2e^{2x}\,dx$.

a. $\frac{e^2-1}{4}$

b. $\frac{e^2}{4}$

c. $\frac{e^2+1}{4}$

d. $e^2 - 1$

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30 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Evaluate $\int e^x \cos(x)\,dx$.

a. $\frac{1}{2}e^x(\cos x + \sin x) + C$

b. $\frac{1}{2}e^x(\cos x - \sin x) + C$

c. $e^x(\sin x + \cos x) + C$

d. $\frac{1}{2}e^x \sin x + C$

31 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Evaluate $\int_1^e \ln(x)\,dx$.

a. 1

b. $e - 1$

c. e

d. 0

32 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Evaluate $\int_0^{\pi} x\sin(2x)\,dx$.

a. $-\pi/2$

b. $\pi/2$

c. 0

d. $-\pi$

33 [Intermediate] • Target: Repeated Integration and Tabular Method

Area between $y = e^x$ and $y = e^{-x}$ from $x = -1$ to $x = 1$:

a. $2(e - 1/e)$

b. $e - 1/e$

c. 0

d. $e + 1/e - 2$

34 [Intermediate] • Target: Looping / Cyclic Integrals and Definite Bounds

Area between $y = x^2$ and $y = x$ equals:

a. 1/3

b. 1/6

c. 1/2

d. 1

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35 [Intermediate] • Target: The integration by parts formula

The region between $y = e^x$ and $y = 1$ from $x = 0$ to $x = 1$ has area:

a. 1

b. $e - 1$

c. e

d. $e - 2$

36 [Intermediate] • Target: Repeated and definite applications

Tabular integration (repeated parts): $\int x^3 e^x \, dx$. Number of IBP steps needed:

a. 2

b. 1

c. 4

d. 3

37 [Intermediate] • Target: Choosing u and dv wisely (LIATE)

Area between $y = f(x)$ and x-axis on $[a, b]$ equals:

a. $\int_a^b |f(x)| \, dx$

b. $\int_a^b f(x) \, dx$ only when $f \geq 0$

c. $f(b) - f(a)$

d. $\int_a^b f(x)^2 \, dx$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The recurring-integral trick • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Reduction formula: $\int \sin^n x \, dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n} \int \sin^{n-2} x \, dx$. Then $\int_0^{\pi/2} \sin^6 x \, dx =$

a. $5\pi/16$

b. $3\pi/16$

c. $5\pi/32$

d. $15\pi/48$

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39 [Challenging] • Target: Definite integrals by parts • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Which formula gives the length of a polar curve $r = f(\theta)$?

a. $\int_{\alpha}^{\beta} r\,d\theta$

b. $\int_{\alpha}^{\beta} |dr/d\theta|\,d\theta$

c. $\int_{\alpha}^{\beta} r^2\,d\theta$

d. $\int_{\alpha}^{\beta} \sqrt{r^2 + (dr/d\theta)^2}\,d\theta$

40 [Challenging] • Target: Reduction formulas • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Trig substitution $\int \sqrt{4 - x^2}\,dx$ with $x = 2\sin\theta$ gives:

a. $\arcsin(x/2) + C$

b. $\frac{x\sqrt{4-x^2}}{2} + 2\arcsin(x/2) + C$

c. $\sqrt{4 - x^2} + C$

d. $-\sqrt{4 - x^2} + C$

41 [Challenging] • Target: Integration by parts with inverse trig functions • Cross-Lesson: 6.01 Related Rates

Centroid $\bar{x}$ of region R with area A:

a. $\frac{a+b}{2}$

b. $\frac{1}{A}\int_a^b f(x)\,dx$

c. $\frac{1}{A}\int_a^b x[f(x) - g(x)]\,dx$

d. $A/2$

42 [Challenging] • Target: A capstone mixed-technique problem • Cross-Lesson: 6.02 Optimization and Curve Sketching

Arc length formula: $L =$

a. $\int_a^b f(x)dx$

b. $\int_a^b \sqrt{1 + [f'(x)]^2}\,dx$

c. $\int_a^b |f'(x)|dx$

d. $\int_a^b [1 + f'(x)]dx$

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43 [Challenging] • Target: Visualizing an area found by integration by parts • Cross-Lesson: 7.01 Integration by Substitution

Reduction formula: $\int \sin^n x\,dx = -\frac{\sin^{n-1} x \cos x}{n} + \frac{n-1}{n}\int \sin^{n-2} x\,dx$. For $n = 4$:

a. $-\sin^3 x \cos x/4 + 3/4 \int \sin^2 x\,dx$

b. $-\sin^3 x \cos x$

c. $\sin^4 x/4$

d. $-\cos x/4$

44 [Challenging] • Target: Choosing between substitution and parts • Cross-Lesson: 8.01 Area Between Curves

Shell method: volume when shell radius $= x$, height $= f(x)$:

a. $2\pi\int_a^b xf(x)dx$

b. $\int_a^b f(x)dx$

c. $\pi\int_a^b x^2 f(x)dx$

d. $\pi\int_a^b f(x)^2 dx$

45 [Challenging] • Target: Standard Products (Polynomial × Exponential / Trig) • Cross-Lesson: 8.02 Volumes of Revolution

Improper integral $\int_0^1 \frac{\ln x}{\sqrt{x}}\,dx$:

a. -4

b. diverges

c. 0

d. -2

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